# Overview of 5G technologies

- 5G is the 5th generation mobile network.
- designed to connect virtually everyone and everything together including machines, objects, and devices.
- Features:
  - deliver higher multi-Gbps peak data speeds
  - ultra low latency
  - more reliability
  - massive network capacity
  - increased availability
  - uniform user experience to more users
  - higher performance
  - improved efficiency

# Invention of 5G

- several companies within the mobile ecosystem that are contributing to bringing 5G to life.
- Qualcomm has played a major role in inventing the many foundational technologies
- 3rd Generation Partnership Project (3GPP) - organization that defines the global specifications for 3G UMTS (including HSPA), 4G LTE, and 5G technologies

# Underlying technologies

- 5G is based on OFDM (Orthogonal frequency-division multiplexing)
- 5G uses 5G NR air interface alongside OFDM principles.
  - A new Air Interface for LTE
  - improve the performance, flexibility, scalability and efficiency of current mobile networks
- 5G also uses wider bandwidth technologies such as sub-6 GHz and mmWave.
  - mmWave refers to higher frequency radio bands ranging from 24GHz to 40GHz
  - Sub-6GHz refers to mid and low-frequency bands under 6GHz.
  - Low-frequency bands are under 1GHz, while mid-bands range from 3.4GHz to 6GHz

# Generations

- First generation - 1G
  - 1980s: 1G delivered analog voice.
- Second generation - 2G
  - Early 1990s: 2G introduced digital voice (e.g. CDMA- Code Division Multiple Access).
- Third generation - 3G
  - Early 2000s: 3G brought mobile data (e.g. CDMA2000).
- Fourth generation - 4G LTE
  - 2010s: 4G LTE ushered in the era of mobile broadband.

# Reasons that 5G will be better than 4G

- 5G is significantly faster than 4G

  - 5G can be significantly faster than 4G, delivering up to 20 Gigabits-per-second (Gbps) peak data rates and 100+ Megabits-per-second (Mbps) average data rates.
- 5G has more capacity than 4G
  - 5G is designed to support a 100x increase in traffic capacity and network efficiency.1
- 5G has significantly lower latency than 4G
  - 5G has significantly lower latency to deliver more instantaneous, real-time access: a 10x decrease in end-to-end latency down to 1ms
- 5G is a unified platform that is more capable than 4G
  - 5G can also natively support all spectrum types (licensed, shared, unlicensed) and bands (low, mid, high), a wide range of deployment models (from traditional macro-cells to hotspots), and new ways to interconnect (such as device-to-device and multi-hop mesh).
- 5G uses spectrum better than 4G
  - 5G is also designed to get the most out of every bit of spectrum across a wide array of available spectrum regulatory paradigms and bands—from low bands below 1 GHz, to mid bands from 1 GHz to 6 GHz, to high bands known as millimeter wave (mmWave).